

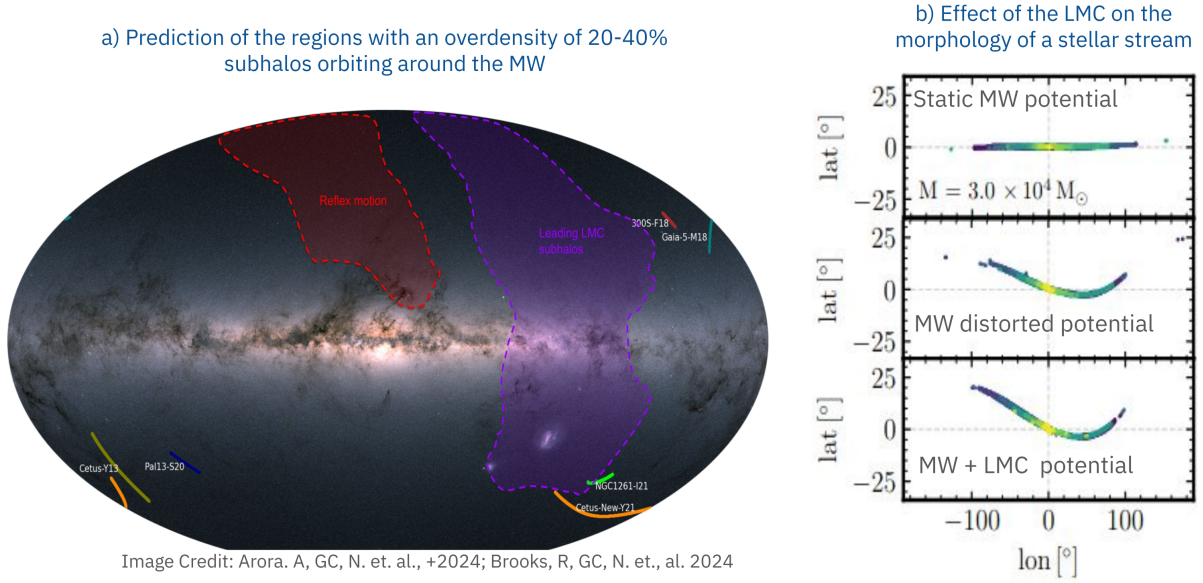


Figure 3: (Left panel) Predictions of regions on the sky with an overdensity of subhalos due to the recent passage from the LMC. (Right panel) Morphology of a stellar stream integrated in a gravitational potential that excludes the presence of LMC (top), of an MW halo that distorts in response to the passage of an LMC (middle), and to a full potential that includes both the MW halo distortions and the LMC itself (bottom).

halo shapes (lead: Dr. Arpit Arora, expected submission early spring 2025); 3) Quantify the impact of wakes and halo distortions on the stellar disk (lead: Dr. Silvio Varela, expected submission early spring 2025). These papers will lay the groundwork for a methodology that showcases the power of BFE for disentangling dynamical processes in cosmological simulations for the first time.

Research Proposal: Constraining the Nature of the Dark Matter particle in the Local Universe at galactic scales

As an NHFP Fellow, I will build simulations to model the response of LG DM halos to external perturbations in CDM and SiDM cosmologies. I will further use these simulations to constrain the properties of the DM particle with key observational signatures that DM halos in disequilibrium produce in stellar halos and in the distribution of satellite galaxies within the LG. To accomplish this, I propose the following three projects:

Project A: Constraining the DM halo response to satellite interactions in CDM and SiDM halos. The density profiles, kinematics, and shapes of cold dark matter (CDM) halos differ from those of self-interacting dark matter (SiDM) halos [e.g., 2]. Consequently, the response of these halos to external perturbations, such as mergers with satellite galaxies, is expected to differ [19]. Interactions between particles in SiDM generally dampen the halo's response. However, there have been no studies directly comparing the responses of CDM and SiDM halos to satellite interactions. Such a study would allow us to compare the observed properties of the MW's halo response to the passage of the LMC, presenting a unique opportunity to constrain the properties of DM in the MW, with broader implications for the fundamental nature of DM.

I will utilize Basis Function Expansions² to characterize the DM halo's response to satellite interactions, similar to my work in [4]. BFEs have been extensively used to

²BFEs are basis function expansions constructed as potential-density pairs that satisfy the Poisson equation. The expansion coefficients fully describe the system's gravitational field.

Characterization of the halo shape of a perturbed halo from a MW-like simulation

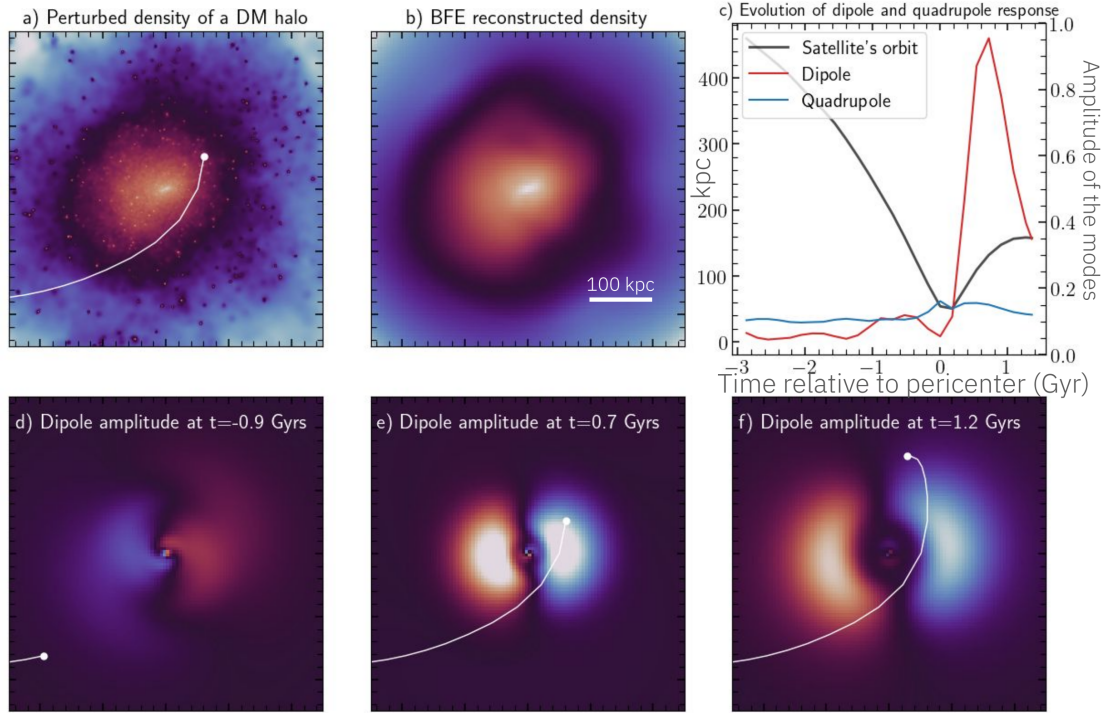


Figure 4: Illustration of the characterization of a perturbed halo from the MWEST suite. The left panel shows the density distribution from the particle data of the simulation, where the halo is lopsided due to the perturbation by the satellite (white line). The top middle panel shows the reconstruction of the density field using Basis Function Expansions (BFEs) with 288 coefficients. The top right panel shows the time evolution of the largest terms in the BFE that characterize the halo response: the *dipole* and the *quadrupole* and their correlation with the satellite’s orbit. The bottom panels show the evolution of the dipole as a function of time, illustrating the rapid response of the halo.

understand the dynamics of DM halos in disequilibrium [20]. Figure 4 shows an example of a DM halo perturbed by an LMC-like satellite (note that the particle data from the satellite was removed to isolate the MW-like DM halo response).

The halo response is primarily captured by a dipole, induced by the satellite near pericenter passages (as shown in Figure 1), affecting the density and kinematics of the halo, along with the dipole response (see Figure 4). I will make specific predictions on the relative differences in the dipole and quadrupole amplitudes in the density, potential, and acceleration fields in CDM and SiDM. These results will provide the first quantification of the MW-LMC response in SiDM and its deviations from CDM, highlighting how BFEs can be effectively used to compare DM halo responses across different DM models.

Deliverables and Legacy: I will publish these results in Fall 2025. I have already developed all necessary methodologies in [4] and through ongoing projects (see Current Research). I will also release code and tutorials to use BFE in cosmological simulations.

Project B: Constraining the dipole response of the MW’s DM halo to the LMC’s passage with satellite galaxies kinematics. As shown in [3, 4] during the ongoing MW-LMC interaction, the MW’s center of mass is pulled by the LMC, thereby inducing a dipolar distribution in the density and velocity of both DM and stellar halos (see Figure 1). Even though there are clear signatures in the data of the density and velocity dipoles (upper panels in Figure 1) that qualitatively agree with our models, we